



Lab 8: VLANs and Inter-VLAN routing

VLANs and Inter-VLAN Routing

A VLAN (Virtual Local Area Network) is a way of splitting a single physical switch into multiple isolated groups. Devices in the same VLAN can communicate freely with each other, but they cannot send traffic to devices in a different VLAN, it is as if they are on completely separate switches, even though they share the same hardware.

Normally, a router creates boundaries between separate networks. VLANs let a switch create those same boundaries in software, without needing extra hardware. Each VLAN is its own broadcast domain, meaning a broadcast sent by a device in VLAN 10, for example, will only reach other devices in VLAN 10.

VLANs are used for:

Security:

Sensitive departments (e.g., HR, Finance) can be kept separate from the rest of the network even if they share the same physical switches.

Reduced broadcast traffic:

Smaller groups mean fewer broadcast frames, which keeps the network fast.

Easier management:

Devices can be grouped by function rather than physical location.

Access Ports and Trunk Ports

There are two types of ports you'll need when configuring VLANs.

Access port:

Connects an end device (PC, printer, etc) to the switch. The port belongs to exactly one VLAN. The device itself does not need to know anything about VLANs.

Trunk port:

Connects two switches together, or a switch to a router. A trunk port carries traffic for multiple VLANs at once. Frames on a trunk link are labelled (tagged) with the VLAN they belong to so the receiving device can tell them apart.

802.1Q Tagging

The 802.1Q standard is how Cisco switches tag VLAN traffic on a trunk. A small label containing the VLAN ID is added to each frame. The receiving device reads this label, strips it, and forwards the frame to the correct VLAN.



Inter-VLAN Routing

Because VLANs are separate broadcast domains, they are also on separate IP subnets. A device in VLAN 10 (e.g., 192.168.10.x) cannot communicate with a device in VLAN 20 (e.g., 192.168.20.x) without going through a router.

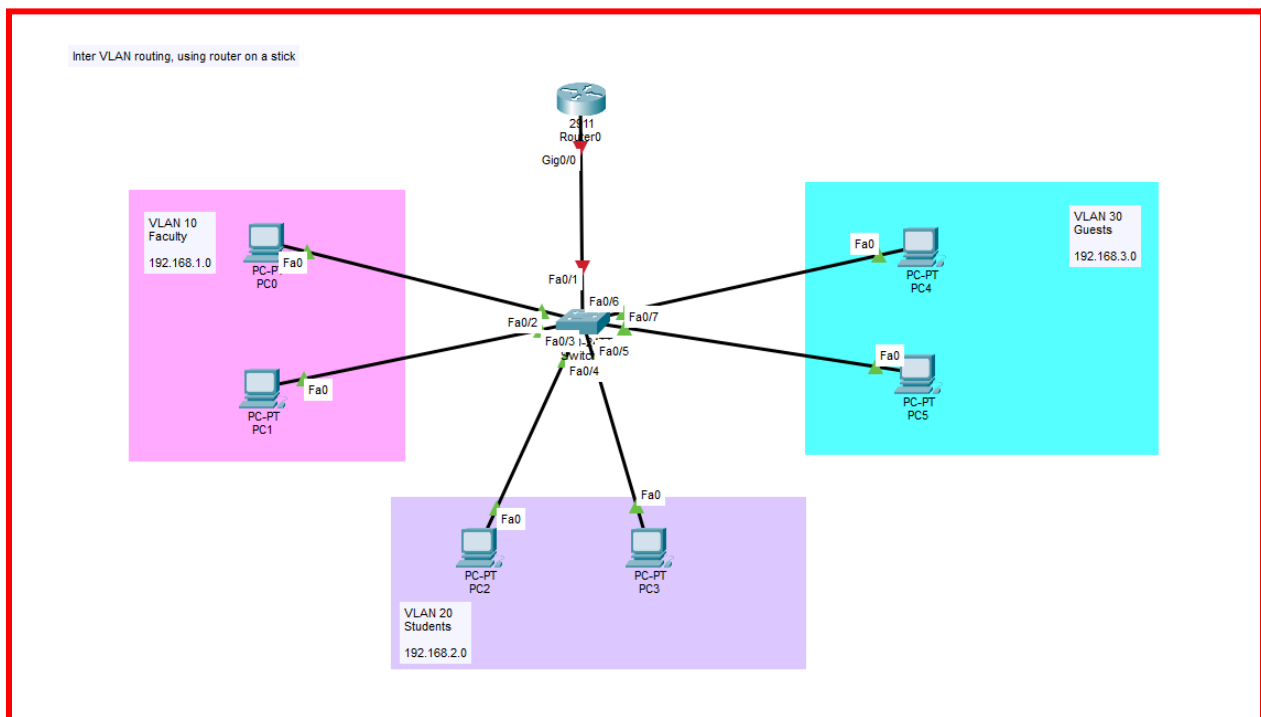
In this lab you will use a technique called router-on-a-stick. A single router interface is divided into sub-interfaces, one for each VLAN. Each sub-interface is tagged to its VLAN and given an IP address that acts as the default gateway for devices in that VLAN. When a host in VLAN 10, for example, wants to reach a host in VLAN 20, it sends the packet to its gateway (the router sub-interface), the router routes it to the VLAN 20 sub-interface, and the switch delivers it to the destination.

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Implementing Inter-VLAN Routing (15 Marks)





The following are the steps and IP addresses you need to follow to observe Inter VLAN Routing on CISCO.

Device Name	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.5	255.255.255.0	192.168.1.1
PC1	192.168.1.6	255.255.255.0	192.168.1.1
PC2	192.168.2.5	255.255.255.0	192.168.2.1
PC3	192.168.2.6	255.255.255.0	192.168.2.1
PC4	192.168.3.5	255.255.255.0	192.168.3.1
PC5	192.168.3.6	255.255.255.0	192.168.3.1

Step 1: Build the Topology

- Add the Switch0 (2960), Router0 (2911) and six PCs. Connect each PC to the switch and the switch to the router using Copper Straight-Through cables. **(ensure you assign the interfaces as they appear in the diagram provided)**



Step 2: Configure VLANs and assign VLAN IDs

- Open **Switch0** and go to the **CLI** tab. Run the following commands.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Faculty
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Students
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Guests
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
-----
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12
Fa0/13, Fa0/14, Fa0/15, Fa0/16
Fa0/17, Fa0/18, Fa0/19, Fa0/20
Fa0/21, Fa0/22, Fa0/23, Fa0/24
Gig0/1, Gig0/2
10 Faculty active
20 Students active
30 Guests active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#int range fa0/6-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#do wr
Building configuration...
```



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```
[OK]
Switch(config)#
Switch#
```

Step 3: Configure trunk ports (needed to enable access to all VLANs) on the link connected to the router

- Open **Switch0** and go to the **CLI** tab. Run the following commands.

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Step 4: Set up IP addresses on the PCs and Test connectivity

- Assign the IP addresses from the details given on the table to each PC (make sure to assign the default gateway too).
- Send a ping from one PC to another PC in different VLANs (e.g; from PC0 to PC5)
- Pings to different VLANs should not work

Step 5: Configure subinterfaces on the router, bind each subinterface to the corresponding VLAN and assign IP address of the subnet

- Open **Router0** and go to the **CLI** tab. Run the following commands (note: if you get a message asking to enter an initial configuration dialog, simply write “no” and press enter).

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int gig0/0.10 //creating a sub-interface for VLAN 10 (you are splitting a physical interface into subinterfaces)
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
```



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```
Router(config-subif)#encapsulation dot1Q 10 //tells the router to use IEEE
802.1Q industry-standard trunking protocol. Enables VLAN tagging (the router
listen to any data packet with a VLAN 10 tag)
Router(config-subif)#ip address 192.168.1.1 255.255.255.0 //the default gateway
of VLAN 10
Router(config-subif)#exit
Router(config)#int gig0/0.20 //creating a sub-interface for VLAN 20 (you are
splitting a physical interface into subinterfaces)
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed
state to up
Router(config-subif)#encapsulation dot1Q 20 //tells the router to use IEEE
802.1Q industry-standard trunking protocol. Enables VLAN tagging (the router
listen to any data packet with a VLAN 20 tag)
Router(config-subif)#ip address 192.168.2.1 255.255.255.0 //the default gateway
of VLAN 20
Router(config-subif)#exit
Router(config)#int gig0/0.30 //creating a sub-interface for VLAN 30 (you are
splitting a physical interface into subinterfaces)
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed
state to up
Router(config-subif)#encapsulation dot1Q 30 //tells the router to use IEEE
802.1Q industry-standard trunking protocol. Enables VLAN tagging (the router
listen to any data packet with a VLAN 30 tag)
Router(config-subif)#ip address 192.168.3.1 255.255.255.0 //the default gateway
of VLAN 30
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Step 6: Test connectivity again

- Send pings (either from the command line or through simple PDU) across the VLANs, it should work (note: it may fail the first time, but will work the next times)

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number.**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1.**
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1. (for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.



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- In case a demo is conducted, there will be deductions based on your answers to the questions asked.

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

